Kassidy Barram

SOFTWARE ENGINEER

Details

Fort Collins
United States
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Links

LinkedIn

Personal GitHub

Skills

Java

Python

C++ / C

JavaScript / TypeScript

Docker

Kubernetes

MongoDB

PyTorch / PySpark

gRPC

Hadoop MapReduce

Apache Spark

Tensor Flow

GitHub

Big Data

Distributed Systems

TCP Networking

Test Driven Development

Machine Learning

Artificial Intelligence

RESTful API

Agile

CI/CD Pipelines

Prometheus

Grafana

Profile

Software engineer with 4.5+ years of experience architecting and deploying distributed systems, decentralized compute infrastructure, and geospatial data pipelines at scale. Designed and launched a live Bittensor subnet supporting 250+ miners and a $500K prize pool, and managed a 200-machine private cloud running 24+ containerized services orchestrated by Kubernetes. Proficient in Python, Java, C++, Docker, Kubernetes, MongoDB, and gRPC. Master's and bachelor's degree in computer science.

Employment History

Software Engineer, qBitTensor Labs, Boulder, CO

DEC 2025 — PRESENT

- Architected a fault-tolerant, distributed validation and scoring system in Python across 17 independent validators and 250+ miners, enforcing incentive mechanisms across a $500K prize pool in a permissionless, trust-less compute economy.
- Developed and maintained public and private miner codebases, providing the foundational framework for participants to submit quantum-inspired solutions to NP-hard optimization problems.
- Engineered a Docker-based sandboxed execution environment for untrusted third-party code, protecting host infrastructure from malicious actors.
- Built and managed data pipelines and persistence layers using MySQL, SQLite, and SQLAlchemy to support solution tracking, scoring, and network state management.

Software Engineer, Project Sustain

OCT 2021 — MAR 2025

- Designed and built 4 spatio-temporal data visualization web applications in React and deck.gl backed by PostGIS and MongoDB, facilitating high-throughput, low-latency geospatial data rendering across distributed environments.
- Built system-wide architecture to continuously deploy, scale, and schedule 24+ distributed services via Kubernetes, eliminating service downtime.
- Single-handedly managed a private cloud totaling 200 machines.
- Developed 2 distributed machine learning model validation services supporting linear regression and classification, executing across 75 machines.

Education

Master of Science, Colorado State University, Fort Collins

AUG 2022 — MAY 2025

- Graduated *Summa cum Laude (4.0 GPA)*
- Awarded Best Paper at 2023 International ACM/IEEE Big Data Conference.
- Published Research in 4 competitive (IEEE/ACM) international computer science conferences.
- Won award at CURC (Celebrating Undergraduate Research and Creativity).

Bachelor of Science, Colorado State University, Fort Collins

AUG 2017 — MAY 2022

- Graduated *Cum Laude (3.95GPA)*
- Awarded National Science Foundation Research Grant to research distributed systems and big data technologies.
- Awarded Pell Grant, and 3 other competitive academic scholarships.
- Participated in research to facilitate communication between Pixhawk 4 Flight Controller and the Inertial Measurement Unit attached to drones using C++.

Internships

Assistant Systems Engineer, CPP Wind Engineering Consultants

MAY 2019 — AUG 2019

- Gathered quantitative data examining the accuracy of multiple data collection methodologies by conducting a speaker test on various pitot-static tubes to quantify frequency response.
- Designed a control station that mixed and regulated the output of tracer gas used in the wind tunnel testing of the air quality from various exhausts produced by buildings.
- Built and assembled city-scape proximity models for data collection in wind tunnels.